EventEcho

Do Indian large-cap stocks move differently around big events, and how sure can we be?

Quantactive Code Conflux · PS-03 · MNNIT Allahabad · [Team name]

7 + NIFTY 50

Large caps across 5 sectors, 2016–2025, one frozen snapshot

21 + 266

Hand-picked macro events plus every quarterly results day

2.7×

Results days move a stock 2.7× as much as normal; placebo 0.9×

8 / 441

Macro CAR tests unusual after Holm correction, mostly Adani

−0.12%

Average macro reaction; BMP, KP and rank tests all p > 0.6

No edge

Trading rule holdout range −0.26 to +0.38 pp includes zero

Rebuilt from one frozen price snapshot in GitHub Actions · **78** integrity checks · **149** unit tests · interactive offline dashboard

From the problem to our approach

Event-study core

Import data
Frozen Yahoo snapshot (SHA-256) + **21** sourced events + **266** results dates

Windows
100-day estimation, **20**-day gap, event windows **±1 / ±3 / ±5**

Returns
Actual vs expected (α + β × NIFTY) → abnormal return

CAR & CAAR
Sum per event; average across events with bootstrap ranges

Visualise
Dashboard: AR/CAR paths, CAAR bands, volatility before/after

Compare
6 event classes, **5** sectors, **7** stocks, results days by stock

Advanced analysis

Regression
OLS market model with Newey–West covariance

Significance
t-test, HAC, Holm over **441** tests, BMP, Kolari–Pynnönen, rank

Window lengths
Same pairs at ±1/±3/±5; count verdict changes

NLP
TF-IDF + logistic regression on **374** real headlines

Unseen sample
Development vs holdout years; placebo days

Extra checks

Market adjustment
Close/Close basis, real NIFTY trading calendar

Confounders
Computed window overlap → **13** independent clusters

Uncertainty
Range on every CAR; cluster bootstrap, **5,000** draws

Trading after costs
3-day rule at **0–30** bps per side, holdout test

Release timing
Release session sourced for **19 / 21** events

Data: how prices, events and dates were collected

2,464

trading sessions, frozen

Prices

Yahoo Finance via yfinance: 7 NSE stocks + ^NSEI

Close for stocks and NIFTY → price returns, dividends excluded on both sides

Stocks re-indexed to real NIFTY sessions; missing prices **never filled**

Raw responses + **SHA-256** saved; every rebuild re-checks it offline

Analysis refuses a snapshot not on the Close/Close basis

21

sourced macro events

Macro events

Monetary **6**, government **5**, earnings **5**, elections **2**, GDP **2**, Hindenburg **1**

Each row: source link, episode ID, timing confidence, Day 0

Day 0 = first session that could react; after-close → next session

Release session sourced for **19 / 21** (e.g. demonetisation reported **20:57 IST** → Day 0 next day)

All 19 agree with our Day 0; **2** unresolved → alternative Day 0 exported

266

results dates, 38 per stock

Results days

Every quarter **Q1 FY17 – Q2 FY26**, a source link on each row

222 board-meeting histories, **43** dated news reports, **1** NSE filing

Seeded random sample vs independent news: **12 / 12** agree; **22** targeted checks agree

3 fixes: a missing quarter, a Sunday date, a double entry

Builder refuses missing/duplicate quarters; gap test 25–150 days

Method: how every number is computed

Expected = α + β × NIFTY | **AR = actual − expected** | **CAR = Σ AR over window** | **CAAR = mean CAR across events**



1 Align

Stocks and NIFTY on the same **2,464** sessions, close-to-close returns

2 Fit

OLS on **100** complete sessions ending **20** before the window

3 AR & CAR

Windows **±1** (3 days), **±3** (7), **±5** (11); blank if a day is missing

4 Pool

CAAR by class, sector, stock; groups with **< 3** clusters get no range

5 Volatility

Realised vol **−20…−1** vs **+1…+20**, annualised √252

Robustness built in

Model re-fit with **60** and **150** sessions: CAR correlation **0.98 / 0.99**

Alternative Day 0 for uncertain events; COVID extreme-day check

Results days: placebo on the same stocks **30** sessions later

Overlap and dependence audit

8 events share trading sessions in their CAR windows

Shared episodes (2022 hike cycle) joined → **13** clusters from **21** events

Overlapping events left out of category summaries; ledger lists caveats

Statistical techniques in detail

t-test with parameter error

$SE^2 = L\sigma^2 + x^T V x$; $t = CAR / SE$

Guards against: Ignoring that α, β are estimated

per CAR

Newey–West (HAC)

Long-run σ^2 with Bartlett weights, bandwidth 4(n/100)^(2/9)

Guards against: Serial correlation in residuals

10 / 441

Holm correction

Sort p; adjusted p(k) = max((m−k+1)p(k)); m = 441

Guards against: Lucky hits (~22 expected by chance)

8 / 441

BMP standardized test

$SCAR_i = CAR_i / SE_i$; $t = \sqrt{N} \cdot mean(SCAR) / sd(SCAR)$

Guards against: Extra volatility on event days

p = 0.92

Kolari–Pynnönen

$t_KP = t_BMP \cdot \sqrt{((1 - \bar{r}) / (1 + (N-1)\bar{r}))}$, $\bar{r}$ = *same-day residual correlation*

Guards against: Stocks moving together on one day

p = 0.93

Corrado / Cowan rank

$U = rank/(T+1) - ½$; $z = \Sigma \bar{U}(t) / (\sqrt{L} \cdot S_U)$

Guards against: Outliers and non-normal returns

p = 0.61

Cluster bootstrap

Resample whole clusters, 5,000 draws, seed 2026, 2.5–97.5 percentiles

Guards against: Overlapping events treated as independent

13 clusters

Variance ratio

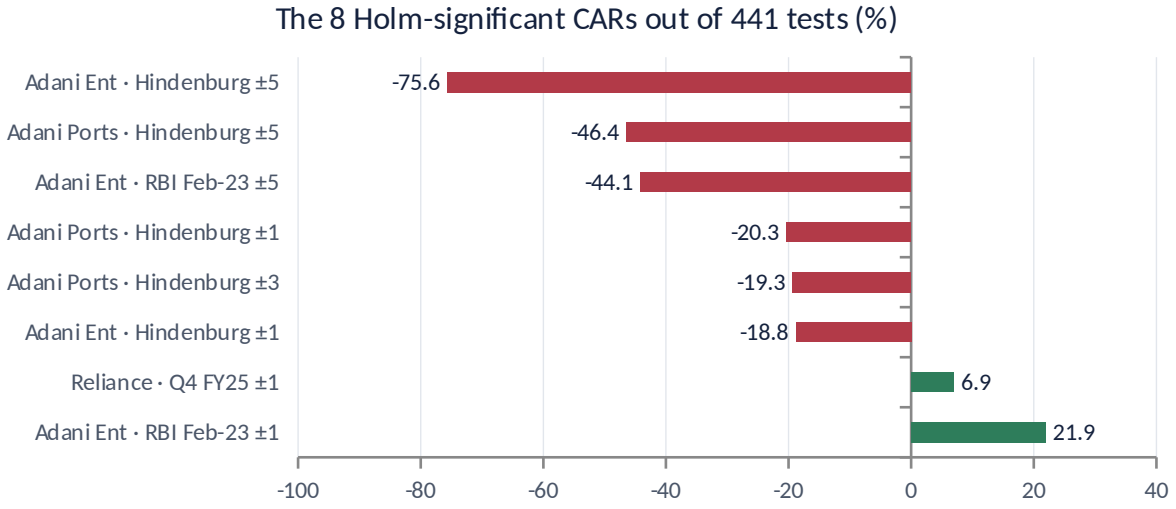
$mean(SCAR^2)$; ≈ *1.02 if the event changes nothing*

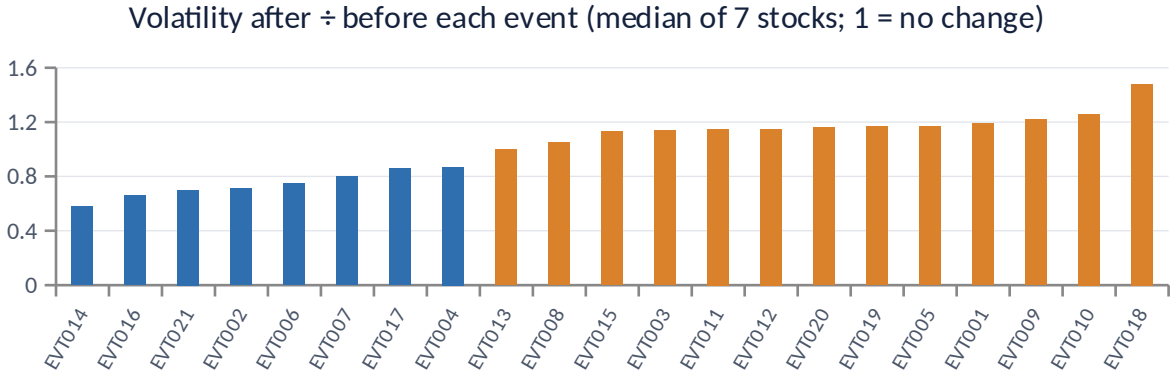
Guards against: Missing a size effect when direction is unknown

2.68 vs 0.88

Also: same tests with **60 / 100 / 150**-day models → **12 / 8 / 11** significant; windows **±1 / ±3 / ±5** → **4 / 1 / 5** (HAC). Code: statistics.py, robust_tests.py, clustered_inference.py, estimation_sensitivity.py.

Macro events: a few huge moves, no reliable average

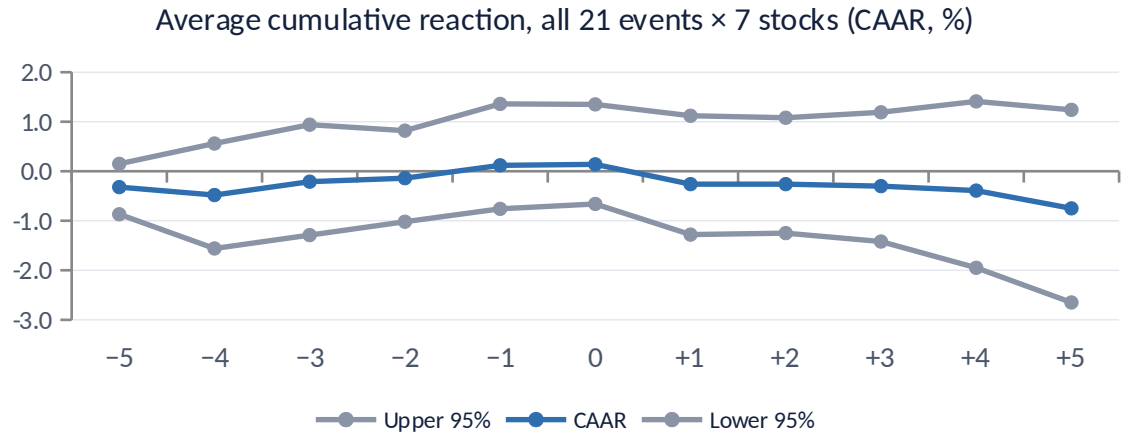




p = 0.92
BMP

p = 0.93
Kolari–Pynnönen

p = 0.61
Rank test



Average CAR, Day −1 to +1	Events / clusters	Mean	95% range
All events	21 / 13	**−0.12%**	−0.53 to +0.49%
Without overlap	13 / 11	**+0.20%**	−0.38 to +0.92%
Government policy	5 / 5	**+0.91%**	−0.09 to +2.14%
Monetary policy	6 / 3	**+0.62%**	+0.03 to +1.77% (rough)

Reading: significant = unusual for the market model, not caused by the event. CAAR band contains **0** on every day; volatility median ratio **1.01** (**52%** of pairs rise, largest GDP May 2024 **1.48×**). Biggest moves are one episode (Adani / Hindenburg) overlapping an RBI decision.

Company results days move stocks 2.7× as much as normal

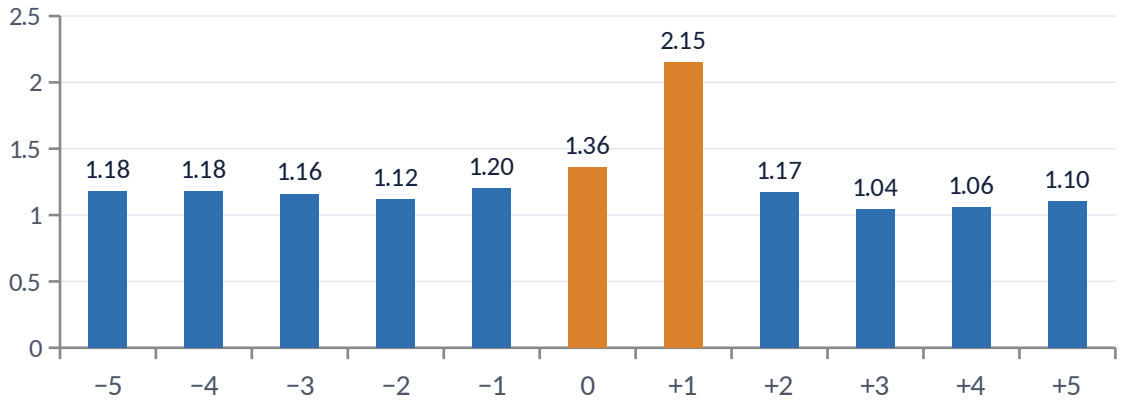
2.68×

results days, range 2.17–3.23

0.88×

placebo days, range 0.70–1.08

Mean |abnormal return| by day around results (%)



Each announcement studied **only for the company that reported** (src/earnings_study.py)

Day +1 peaks (**4.8×** normal) → many results arrive after the close or on Saturday

19% of results days have |SCAR| > 1.96 vs **5%** expected; **49** clusters in the bootstrap

Sample, Day −1 to +1	N	Ratio	95% range
All results days	266	**2.68**	2.17 – 3.23
Without macro overlap	217	**2.55**	2.00 – 3.15
2016–2022 (development)	182	**2.15**	1.69 – 2.67
2023–2025 (holdout)	84	**3.82**	2.72 – 4.95
Placebo: ordinary days	264	**0.88**	0.70 – 1.08

Stock	Ratio	Both periods?
Infosys	4.56	Yes
TCS	2.53	Yes
HDFC Bank	3.08	Early only
Reliance	2.44	Late only

Stock	Ratio	Both periods?
ITC	2.46	No
Adani Ports	2.55	No
Adani Ent.	1.15	No
Signed mean	−0.28%	p = 0.13

Not claimed: a sector ranking (IT **3.5** vs Banking **3.1**, ranges overlap) or any direction of the move. Effect fades in wider windows: **1.90** at ±3, **1.43** at ±5.

Unseen events, trading rule and NLP classifier

Do macro patterns repeat later?

3 of 6

checks repeat at Day −1 to +1

Check (±1)	Dev	Hold	Rep.
Overall direction	+0.18	−0.35	No
Elections	−1.62	−1.28	Yes
Govt policy	+1.14	+0.58	Yes
Monetary	+1.07	−0.27	No

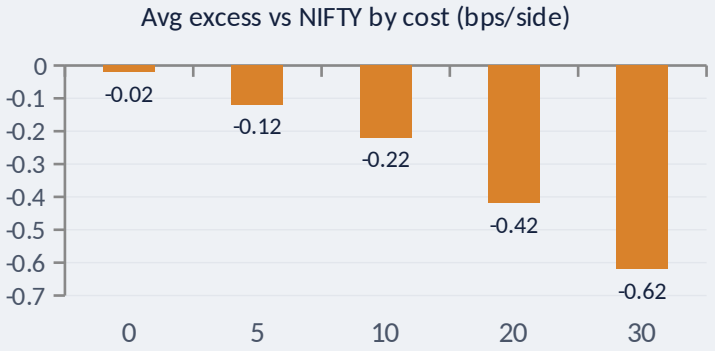
Sign-guessing from class: **80%** on 5 classes, not beyond chance

Strong version: results days **182 vs 84** → effect holds

Hypothetical trading rule

−0.26 to +0.38

pp holdout 95% range — includes zero



Buy at close of first session after Day 0, hold **3** sessions, equal weight

trading.py simulates; **8** holdout clusters bootstrapped

NLP headline classifier

81% model

29% majority guess

TF-IDF (words 1–2, chars 3–5) + **logistic regression**, settings fixed before testing

374 real sourced headlines, **6** classes, **49** search groups

Grouped CV holds out whole groups → **75%** (majority **23%**)

Tested on **21** registry headlines never trained on

Flags **4** disagreements; never changes a return

Code, pipeline and audit on GitHub

Acquire → yfinance → raw + SHA-256

Freeze → snapshot in CI artifact

Analyse → run_analysis.py offline

Dashboard → build_demo.py → HTML

Verify → verify_release.py · 78 checks

Publish → commit to branch → PR

lordabhishek10242/EventEcho

```
├─ .github/workflows/  tests · historical-rebuild · recover-snapshot
├─ src/  19 modules, ~4,600 lines
│  ├─ data.py · snapshots.py      prices, alignment, SHA-256
│  ├─ events.py · event_audit.py   Day 0, overlap audit
│  ├─ event_study.py · statistics.py model, CAR, t/HAC/Holm
│  ├─ robust_tests.py · pooled_analysis.py BMP/KP/rank, CAAR
│  ├─ earnings_study.py   266 results days, placebo
│  ├─ volatility.py · sensitivity.py · trading.py
│  ├─ headline_classifier.py   NLP model + baselines
│  └─ reporting.py   dashboard + key findings
├─ scripts/  acquire · rebuild · run_analysis · build_demo · verify
├─ tests/  26 files · 149 network-free tests
├─ results/  event_master · earnings_events · headline_corpus
│  └─ reproduced/  40+ CSVs · manifest · validation · dashboard
└─ docs/  audit · release times · earnings · judge Q&A
```

78 integrity checks

149 unit tests

3 CI workflows

What the 78 checks verify

CAR = Σ AR; Holm p-values recompute over **441** tests

Dashboard tables **equal** the shipped CSVs

Split order, cost arithmetic, overlap symmetry

Variance ratio and SCARs recompute; calendar checksum matches

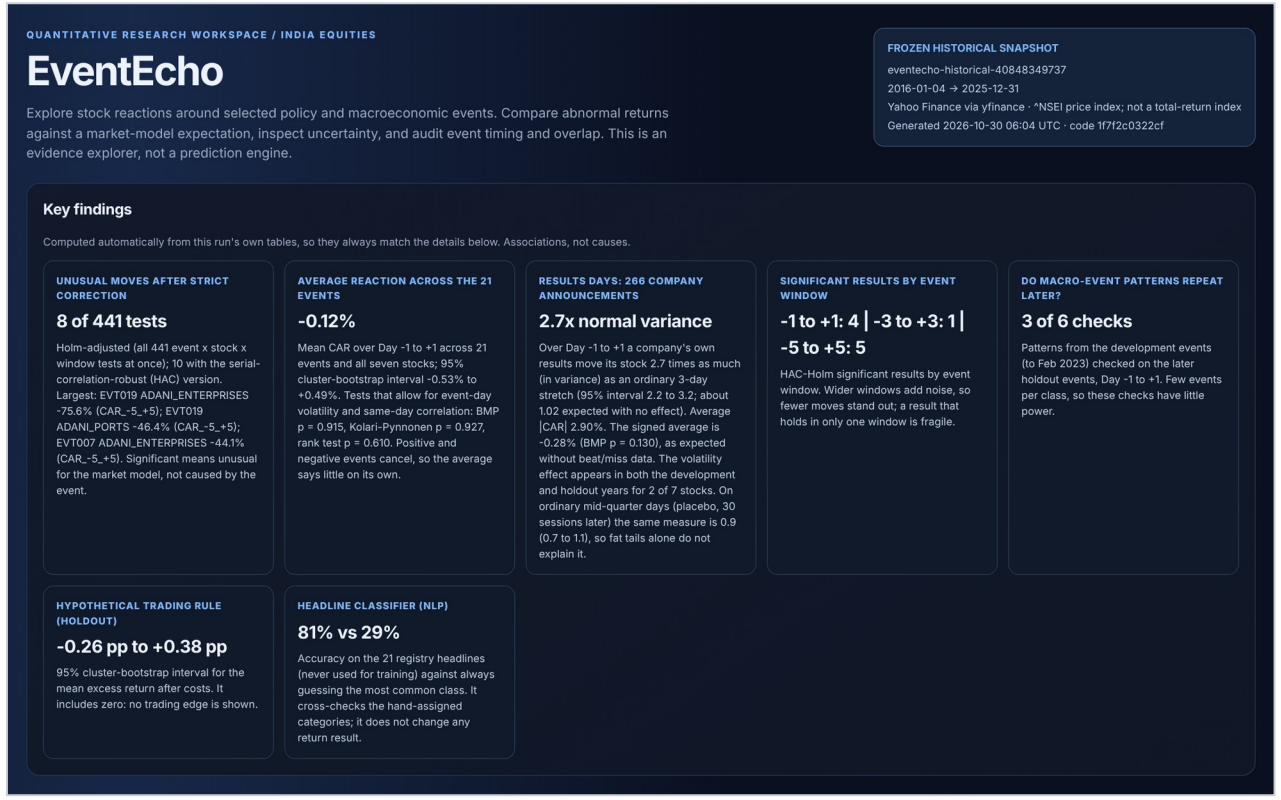
Audit trail

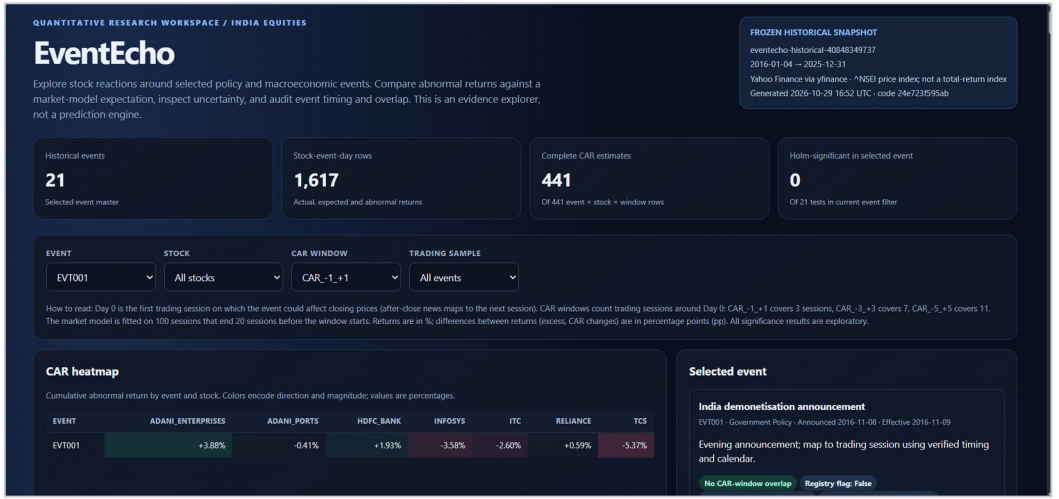
manifest.json: code commit, snapshot ID, price SHA-256, settings

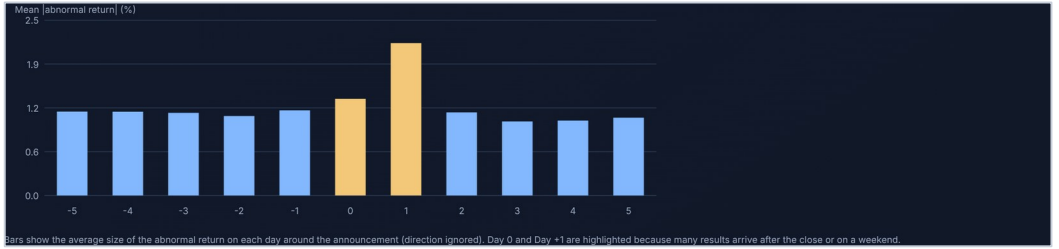
Docs: evidence audit, release-time check, earnings write-up

PRs **#37–#38** merged, **#39** in review; cited run **41011773210**

Interactive dashboard and conclusion







Filters for event, stock, window and sample · CAR heatmap · CAAR bands · robust tests · results-day panel · overlap and timing audit · provenance. **One offline HTML file.**

Results days move stocks 2.7× · no reliable macro average · no trading edge

Limits we state: 21 hand-picked events, no beat/miss data, daily closes, one price vendor. Next: analyst estimates, filing timestamps, more stocks. Questions?